Tobias Riedlinger

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Scholar Tobias Riedlinger | Citations: 84, h-index: 4
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Research Interests

Mathematical Theory of Deep Neural Networks
Statistical Learning Theory for Deep Learning
Generative Models and Artificial Data
Uncertainty Quantification in Machine Learning
Reliable Artificial Intelligence

Academic Positions

Since 2023 **Postdoctoral Researcher**, *TU Berlin*.
2019–2023 **Doctoral Researcher**, *University of Wuppertal*.

Academic Education

2023 **PhD in Mathematics**, *University of Wuppertal*, Summa Cum Laude.
Methods and Applications of Uncertainty Quantification for Object Recognition
Supervisors: Prof. Dr. Hanno Gottschalk, Prof. Dr. Matthias Rottmann

2019 **Master of Science Physics**, *University of Tübingen, Germany*, 1,0.
Extension of Dynamics at Spacetime Singularities
Supervisors: Prof. Dr. Roderich Tumulka, Prof. Dr. Frank Loose

2018 **Bachelor of Science Physics**, *University of Tübingen, Germany*, 1,4.
The Dirac Equation and Singular Spacetimes
Supervisor: Prof. Dr. Roderich Tumulka

List of Publications

* equal contribution, † corresponding author

Selected Publications

[1] Tobias J Riedlinger*, Kira Maag*, and Hanno Gottschalk. *Towards Reliable Detection of Empty Space: Conditional Marked Point Processes for Object Detection*. arXiv preprint arXiv:2506.21486 (2025) — accepted as full paper at ICLR 2026

[2] Tobias Riedlinger*†, Marius Schubert, Sarina Penquitt*, Jan-Marcel Kezmann, Pascal Colling, Karsten Kahl, Lutz Roese-Koerner, Michael Arnold, Urs Zimmermann, and Matthias Rottmann. *Lmd: Light-weight prediction quality estimation for object detection in lidar point clouds*. International Journal of Computer Vision (2025), pages 4349–4365

[3] Tobias Riedlinger†, Matthias Rottmann, Marius Schubert, and Hanno Gottschalk. *Gradient-Based Quantification of Epistemic Uncertainty for Deep Object Detectors*. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision. 2023, pages 3921–3931

Journal Articles

[4] Tobias Riedlinger*† and Kira Maag*. *Advanced Analysis of Pixel-Wise Gradient Uncertainty for Convolutional Neural Networks*. In: Computer Vision, Imaging and Computer Graphics Theory and Applications. Edited by T. Bashford-Rogers, D. Meneveaux, M. Ziat, M. Ammi, S. Jänicke, H. Purchase, P. Radeva, A. Furnari, K. Bouatouch, and A. A. de Sousa. Cham: Springer Nature Switzerland, 2026, pages 239–258

[2] Tobias Riedlinger*†, Marius Schubert, Sarina Penquitt*, Jan-Marcel Kezmann, Pascal Colling, Karsten Kahl, Lutz Roese-Koerner, Michael Arnold, Urs Zimmermann, and Matthias Rottmann. *Lmd: Light-weight prediction quality estimation for object detection in lidar point clouds*. International Journal of Computer Vision (2025), pages 4349–4365

Top-Tier Conference Papers

[1] Tobias J Riedlinger*, Kira Maag*, and Hanno Gottschalk. *Towards Reliable Detection of Empty Space: Conditional Marked Point Processes for Object Detection*. arXiv preprint arXiv:2506.21486 (2025) — accepted as full paper at ICLR 2026

[3] Tobias Riedlinger†, Matthias Rottmann, Marius Schubert, and Hanno Gottschalk. *Gradient-Based Quantification of Epistemic Uncertainty for Deep Object Detectors*. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision. 2023, pages 3921–3931

Conference Papers

[5] Kira Maag* and Tobias Riedlinger*. *Pixel-Wise Gradient Uncertainty for Convolutional Neural Networks Applied to Out-of-Distribution Segmentation*. In: Proceedings of the 19th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications - Volume 2: VISAPP. INSTICC. SciTePress, 2024, pages 112–122

[6] Marius Schubert*, Tobias Riedlinger*, Karsten Kahl, and Matthias Rottmann. *Deep Active Learning with Noisy Oracle in Object Detection*. In: Proceedings of the 19th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications - Volume 2: VISAPP. INSTICC. SciTePress, 2024, pages 375–384

[7] Tobias Riedlinger*, Marius Schubert*, Karsten Kahl, Hanno Gottschalk, and Matthias Rottmann. *Towards Rapid Prototyping and Comparability in Active Learning for Deep Object Detection*. In: Proceedings of the 19th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications - Volume 2: VISAPP. INSTICC. SciTePress, 2024, pages 366–374

[8] Marius Schubert, Tobias Riedlinger, Karsten Kahl, Daniel Kröll, Sebastian Schoenen, Siniša Šegvić, and Matthias Rottmann. *Identifying label errors in object detection datasets by loss inspection*. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision. 2024, pages 4582–4591

[9] Tobias Riedlinger*†, Marius Schubert, Sarina Penquitt*, Jan-Marcel Kezmann, Pascal Colling, Karsten Kahl, Lutz Roese-Koerner, Michael Arnold, Urs Zimmermann, and Matthias Rottmann. *LMD: Light-weight Prediction Quality Estimation for Object Detection in Lidar Point Clouds*. In: Pattern Recognition. Edited by U. Köthe and C. Rother. Cham: Springer Nature Switzerland, 2024, pages 85–99

Book Chapters

[10] Tobias Riedlinger†, Marius Schubert, Karsten Kahl, and Matthias Rottmann. *Uncertainty Quantification for Object Detection: Output- and Gradient-Based Approaches*. In: Deep Neural Networks and Data for Automated Driving: Robustness, Uncertainty Quantification, and Insights towards Safety. Edited by T. Fingscheidt, H. Gottschalk, and S. Houben. Cham: Springer International Publishing, 2022, pages 251–275

Under Review

[11] Jonathan Klees, Tobias Riedlinger, Peter Stehr, Bennet Böddecker, Daniel Kondermann, and Matthias Rottmann. *Probabilistic Label Spreading: Efficient and Consistent Estimation of Soft Labels with Epistemic Uncertainty on Graphs*. 2026. arXiv: 2602.04574 [cs.LG]. URL: https://arxiv.org/abs/2602.04574

[12] Sarina Penquitt, Tobias Riedlinger, Timo Heller, Markus Reischl, and Matthias Rottmann. *Learning to Detect Label Errors by Making Them: A Method for Segmentation and Object Detection Datasets*. 2025. arXiv: 2508.17930 [cs.LG]. URL: https://arxiv.org/abs/2508.17930

[13] Hanno Gottschalk, Emil Partow, and Tobias J Riedlinger. *Consistency of Learned Sparse Grid Quadrature Rules using NeuralODEs*. arXiv preprint arXiv:2507.01533 (2025)

[14] Emily C Ehrhardt, Hanno Gottschalk, and Tobias J Riedlinger. *Numerical and statistical analysis of NeuralODE with Runge-Kutta time integration*. arXiv preprint arXiv:2503.10729 (2025)

In Preparation

[15] Hanno Gottschalk and Tobias Riedlinger. *Regularity of Solutions to Beckmann's Parametric Optimal Transport*. 2026

[16] Leroy Odunlami, Tobias Riedlinger, and Hanno Gottschalk. *Neural Networks might fail to learn from Synthetic Data*. 2026

Conference Presentations (selected)

2026 GAMM (96th Annual Meeting of the International Association of Applied Mathematics and Mechanics), Stuttgart Germany
(contributed talk: *Hölder Regularity of Solutions to the Beckmann Problem with Quadratic Cost* [15])

2024 VISAPP (19th International Conference on Computer Vision Theory and Applications), Rome Italy
(oral presentation: *Pixel-Wise Gradient Uncertainty for Convolutional Neural Networks Applied to Out-of-Distribution Segmentation* [5])

2023 WACV (IEEE/CVF Winter Conference on Applications of Computer Vision), Waikoloa Hawaii
(oral presentation: *Gradient-Based Quantification of Epistemic Uncertainty for Deep Object Detectors* [3])

2022 Robust Understanding of Stree Scenes Using Computer Vision Workshop, Zagreb Croatia
(*Gradient-Based Quantification of Epistemic Uncertainty for Deep Object Detectors*)

GAMM (92nd Annual Meeting of the International Association of Applied Mathematics and Mechanics), Aachen Germany
(contributed talk: *Input Reconstruction from Compressed Latent Features in Deep Neural Networks*)

Research Visits

2025 Research Visit (1 week), Faculty of Electrical Engineering and Computing University of Zagreb, Croatia
Host: Anja Delić, In-person collaboration on out-of-distribution dataset

2024 Research Visit (1 week), Faculty of Electrical Engineering and Computing University of Zagreb, Croatia
Host: Siniša Šegvić, Seminar Talk: *Pixel-Wise Gradient Uncertainty for Convolutional Neural Networks Applied to Out-of-Distribution Segmentation* [5]

Student Supervision

PhD Co-Supervision

since 2024 Quality Assessment of Generative Models and Synthetic Data, *L. Odunlami*, TU Berlin.
(with Hanno Gottschalk)

Master's Theses

2026 Information Geometry and Singular Learning Theory for ReLu Neural Networks, *L. Papp*.

Studies on the Regularity Theory of the Flow Matching Vector Field, *F. Ermel*.

Conditional Marked Poisson Point Process Networks for Synthetic Radar Point Cloud Generation, *L. Huber*.

Object Detection and Region Proposal based on Poisson Point Processes, *W. Chang*.

2025 Self-Consuming Generative Models and Random Walks in Parameter Estimation, *W. Wick*.

2024 Error Analysis for Learned Sparse Grid Quadrature Rules Generated with Flow Matching, *E. Partow*.

Moduli Spaces for Shallow Neural Networks and the CLT, *B. Richter*.

Generative Approaches to Input Feature Reconstruction of Deep Neural Networks for Image Classification, *J. Klees*.

Bachelor's Theses

2026 Symmetries and Moduli Spaces of Deep Neural Networks, *T. Nguyen*.

2025 Generative Learning with Beckmann Networks, *J. Güsten*.

2024 Statistical Learning Theory for NeuralODE, *E. Ehrhardt*.

Variational Representations of Divergences and Adversarial Learning, *P. López*.

Teaching and Mentoring Experience

Courses Taught (Instructor/Organizer)

2026 Mathematical Perspectives on Machine Learning Algorithms (Seminar), *TU Berlin*.

2025 Uncertainty Quantification in Machine Learning (Seminar), *TU Berlin*.

Teaching Assistance

2025 Mathematical Foundations of Machine Learning 2, *TU Berlin*.

Mathematical Foundations of Machine Learning, *TU Berlin*.

2024 Numerical Linear Algebra, *TU Berlin*.

Mathematik für Physikerinnen und Physiker 2, *TU Berlin*.
(Mathematics for Physicists)

2023 Mathematical Foundations of Machine Learning 2, *TU Berlin*.

2022 High-Dimensional Probability Theory, *University of Wuppertal*.

2021 High-Dimensional Probability Theory, *University of Wuppertal*.

2020 Versicherungsmathematik, *University of Wuppertal*.
(Mathematics of Insurance)

2019 Mathematical Statistical Physics, *University of Tübingen*.

Gewöhnliche Differentialgleichungen, *University of Tübingen*.
(Ordinary Differential Equations)

Academic Service

2025 Referee/Program Committee, *AAAI 2026*, Singapore.

2024 Referee, *BMVC 2024*, Glasgow.
(Robust Recognition in the Open World Workshop)

Referee + Session Chair, *VISAPP 2024*, Rome.